LIU HE

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Department of Electronic Engineering, Tsinghua University, Beijing, China

RESEARCH INTERESTS

Architecture Design with Advanced and Emerging Computing Technologies

- 3D-integrated heterogeneous memory architecture design for large language model acceleration.

Agent-Driven Computer Architecture Design

- Kernel Agent design for high-performance operator optimization.

EDUCATION

Tsinghua University, Beijing, China **09/2023 - Present** Ph.D. - Electronic Science and Technology
Advisor: Associate Professor Hongyang Jia

Tsinghua University, Beijing, China **08/2019 - 06/2023** B.Eng. - Electronic Engineering

Ningbo Xiaoshi High School, Ningbo, Zhejiang, China **06/2016 - 05/2019**

Ningbo Haishu Foreign Language School, Ningbo, Zhejiang, China **09/2013 - 06/2016**

WORK EXPERIENCE

Intel Labs China, Beijing, China
Research intern, General-purpose processor branch prediction unit design **06/2022 - 12/2022**

Biren Technology, Shanghai, China
Intern, Kernel agent design **06/2025 - 07/2025**

PUBLICATIONS

Conferences:

1. **L. He**, F. Zhou, C. Peng, S. Dong, Z. Zhang, H. Yang, Y. Liu, G. Zhang, H. Jia, "SHyLA: 3D-Stacked NVM-DRAM Hybrid LLM-Inference Architecture Exploiting Data and Memory Heterogeneity," *2026 IEEE/ACM 53rd Annual International Symposium on Computer Architecture (ISCA)*, Raleigh, NC, USA, June 2026.
2. **L. He**, Y. Wang, Z. Huang, S. Fan, C. Tang, S. Zhang, L. Lei, H. Yang, Y. Liu, H. Jia, "Exploring Approximation and Dataflow Co-Optimization for Scalable Transformer Inference Architecture on the Edge," *2024 IEEE 37th International System-on-Chip Conference (SOCC)*, Dresden, Germany, September 2024.
3. Y. Wang, S. Dong, Z. Huang, Y. You, **L. He**, H. Yang, Y. Liu, H. Jia, "HyC-LoRA: Memory Efficient LoRA Fine-tuning with Hybrid Activation Compression," *2025 8th Conference on Machine Learning and Systems (MLSys)*, Santa Clara, CA, USA, May 2025.

Journal:

1. **L. He**, Z. Huang, Y. Wang, S. Fan, C. Tang, S. Zhang, L. Lei, H. Yang, Y. Liu, H. Jia, "SASDenSebLE: A Compact Vision Transformer Inference Architecture With Saturation-Approximate Softmax Dataflow Enabling Sequence-Parallelism Boosted Layer-Fusion Execution," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.

AWARDS

Comprehensive Excellence Scholarship for Graduate Students - Tsinghua University	2025
Comprehensive Excellence Scholarship for Undergraduate Students - Tsinghua University	2022
Comprehensive Excellence Scholarship for Undergraduate Students - Tsinghua University	2021

Academic Excellence Scholarship for Undergraduate Students - Tsinghua University 2020

SHORT BIO

Liu He is currently a third-year Ph.D. student in the Department of Electronic Engineering at Tsinghua University, Beijing, China. She received her B.Eng. degree in Electronic Engineering from Tsinghua University in 2023. Her research focuses on computer architecture, with current interests in kernel agent design for high-performance operator optimization and 3D-integrated heterogeneous memory architecture design for large language model inference acceleration.

She has research experience in large language model inference, agent workflow design, operator optimization, and hardware/system co-evaluation. She is familiar with Verilog, C++, and Python, and has experience with LLM architectures, AI agent design, Claude Code, Codex, and simulators such as gem5 and GPGPU-Sim.